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TREND-BASED MOCK PAPER

GATE Architecture and Planning

Mock Paper — 01

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Architecture and Planning
Paper:	Mock 01
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

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Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

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Section A: General Aptitude

Q1. [1-NAT] [1 mark]

Scale 1:100, drawing 10cm -> actual:

(Numerical Answer Type — Enter the numerical value)

Section B: Architecture and Planning

Q2. [1-MCQ] [1 mark]

Gauss-Seidel method converges when the coefficient matrix is:

- (A) Symmetric
 - (B) Positive definite
 - (C) Diagonally dominant
 - (D) Non-singular
-

Q3. [1-MCQ] [1 mark]

If a 3x3 matrix has eigenvalues 1, 2, and 3, then $\text{trace}(A^2)$ equals:

- (A) 14
 - (B) 6
 - (C) 36
 - (D) 12
-

Q4. [1-MCQ] [1 mark]

$P(|Z| \leq 1)$ for $Z \sim N(0,1)$ is approximately:

- (A) 0.68
- (B) 0.32
- (C) 0.95
- (D) 0.16

Q5. [1-MCQ] [1 mark]

Number of spanning trees in complete graph K_4 :

- (A) 4
- (B) 8
- (C) 16
- (D) 12

Q6. [1-NAT] [1 mark]

If $f(x) = x^3 - 3x + 2$, number of real roots:

(Numerical Answer Type — Enter the numerical value)

Q7. [1-MCQ] [1 mark]

Laplace transform of $e^{(-3t)}u(t)$ is:

- (A) $1/(s+3)$
- (B) $1/(s-3)$
- (C) $1/s$
- (D) e^{-3}/s

Q8. [1-MCQ] [1 mark]

For the ODE $dy/dx = y/x + \sqrt{y^2/x^2}$, with $y(1)=0$, the solution is:

- (A) $y = 0$
- (B) $y = x \ln(x)$
- (C) $y = x \sin(\ln x)$
- (D) $y = x^2$

Q9. [1-MCQ] [1 mark]

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$ equals:

- (A) e
- (B) e^2
- (C) 1
- (D) infinity

Q10. [1-MCQ] [1 mark]

Consider $f(x) = |x|$ on $[-1,1]$. This function is:

- (A) Differentiable everywhere
- (B) Continuous but not differentiable at 0
- (C) Neither continuous nor differentiable
- (D) Differentiable at 0 only

Q11. [1-NAT] [1 mark]

The area of a circle of radius r is:

(Numerical Answer Type — Enter the numerical value)

Q12. [1-NAT] [1 mark]

Fourier series of $f(x)=x$ on $(-\pi,\pi)$ has only:

(Numerical Answer Type — Enter the numerical value)

Q13. [1-NAT] [1 mark]

A random variable $X \sim \text{Poisson}(\lambda=4)$. $P(X \geq 2)$ is approximately:

(Numerical Answer Type — Enter the numerical value)

Q14. [2-MCQ] [2 marks]

If $f(z) = z^2 + 3z + 2$ is analytic, then $\operatorname{Re}(f(i)) =$

- (A) -1
- (B) -2
- (C) 1
- (D) 2

Q15. [2-MCQ] [2 marks]

Solution of $y'' + 4y = 0$ with $y(0)=1$, $y'(0)=0$, $y(\pi/2)$ equals:

- (A) 1
- (B) 0
- (C) -1
- (D) $\cos(\pi)$

Q16. [2-MCQ] [2 marks]

Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq$:

- (A) $\|u\| * \|v\|$
- (B) $\|u\| + \|v\|$
- (C) $\|u\| / \|v\|$
- (D) $\sqrt{\|u\| * \|v\|}$

Q17. [2-NAT] [2 marks]

The differential equation $y \, dx - x \, dy = 0$ has solution:

(Numerical Answer Type — Enter the numerical value)

Q18. [2-MSQ] [2 marks]

For a Poisson distribution with mean 4, variance is:

- (A) 4
- (B) 2
- (C) 16
- (D) 1

Q19. [2-MCQ] [2 marks]

The rank of the matrix $\begin{bmatrix} 1, 2, 3 \\ 2, 4, 6 \\ 3, 6, 9 \end{bmatrix}$ is:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q20. [2-MCQ] [2 marks]

The vector $(1, 2, 3)$ projected onto $(2, 1, 0)$ is:

- (A) $(4/5, 2/5, 0)$
- (B) $(8/5, 4/5, 0)$
- (C) $(2, 1, 0)$
- (D) $(1, 2, 3)$

Q21. [2-MSQ] [2 marks]

The Laplace transform of $t^2 e^{-2t}$ is:

- (A) $1/(s+2)^2$
- (B) $1/(s-2)^2$
- (C) $s/(s+2)^2$

(D) $2/(s+2)^2$

Q22. [2-MCQ] [2 marks]

Rolle's theorem requires $f(x)$ to be:

- (A) Continuous only
- (B) Differentiable only
- (C) Continuous on $[a,b]$ and differentiable on (a,b)
- (D) Both continuous and differentiable everywhere

Q23. [2-MCQ] [2 marks]

$d/dx[\ln(\sec x) + \ln(\tan x)]$ evaluated at $x = \pi/4$ equals:

- (A) 0
- (B) 2
- (C) 4
- (D) 1

Q24. [2-MCQ] [2 marks]

Which of the following is a probability generating function for Poisson(λ)?

- (A) $e^{\lambda(z-1)}$
- (B) $1/(1-z)$
- (C) $\ln(1-z)$
- (D) $e^{-\lambda}(1+z)$

Q25. [2-MCQ] [2 marks]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

- (A) 0.2
- (B) 0.5
- (C) 0.6
- (D) 0.4

Q26. [2-MSQ] [2 marks]

For matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $A^{50} \cdot \begin{bmatrix} 1 & 0 \end{bmatrix}^T$ equals:

- (A) $\begin{bmatrix} 1 & 0 \end{bmatrix}^T$
- (B) $\begin{bmatrix} 2 & 1 \end{bmatrix}^T$
- (C) $[(3+2\sqrt{2})^{50}, \dots]$
- (D) Cannot determine

Q27. [2-MCQ] [2 marks]

Fourier series of $f(x)=x$ on $(-\pi, \pi)$ has only:

- (A) Sine terms
- (B) Cosine terms
- (C) Both
- (D) Neither

Q28. [2-MCQ] [2 marks]

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$ equals:

- (A) e
- (B) e^2
- (C) 1
- (D) infinity

Q29. [2-MSQ] [2 marks]

A random variable $X \sim \text{Poisson}(\lambda=4)$. $P(X \geq 2)$ is approximately:

- (A) 0.908
- (B) 0.092
- (C) 0.762
- (D) 0.195

Q30. [2-MSQ] [2 marks]

A random variable $X \sim \text{Poisson}(\lambda=4)$. $P(X \geq 2)$ is approximately:

- (A) 0.908
- (B) 0.092
- (C) 0.762
- (D) 0.195

Q31. [2-NAT] [2 marks]

The rank of the matrix $\begin{bmatrix} 1, 2, 3 \\ 2, 4, 6 \\ 3, 6, 9 \end{bmatrix}$ is:

(Numerical Answer Type — Enter the numerical value)

Q32. [2-MCQ] [2 marks]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

- (A) 0.2
- (B) 0.5
- (C) 0.6
- (D) 0.4

Q33. [2-MCQ] [2 marks]

If $f(x) = x^3 - 3x + 2$, number of real roots:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q34. [1-MCQ] [1 mark]

Taj Mahal built by:

- (A) Akbar
- (B) Shah Jahan
- (C) Aurangzeb
- (D) Jahangir

Q35. [1-MCQ] [1 mark]

Kath-Kuni from:

- (A) Rajasthan
- (B) Himachal Pradesh
- (C) Kerala
- (D) Gujarat

Q36. [1-MCQ] [1 mark]

Taj Mahal built by:

- (A) Akbar
- (B) Shah Jahan
- (C) Aurangzeb

(D) Humayun

Q37. [1-MCQ] [1 mark]

M20 concrete mix ratio:

- (A) 1:1.5:3
- (B) 1:2:4
- (C) 1:3:6
- (D) 1:1:2

Q38. [2-MCQ] [2 marks]

M20 concrete mix ratio:

- (A) 1:1.5:3
- (B) 1:2:4
- (C) 1:3:6
- (D) 1:1:2

Q39. [2-MCQ] [2 marks]

Taj Mahal built by:

- (A) Akbar
- (B) Shah Jahan
- (C) Aurangzeb
- (D) Humayun

Q40. [2-NAT] [2 marks]

Taj Mahal built by:

(Numerical Answer Type — Enter the numerical value)

Q41. [2-MCQ] [2 marks]

M20 concrete mix ratio:

- (A) 1:1.5:3
- (B) 1:2:4
- (C) 1:3:6
- (D) 1:1:2

Q42. [2-MCQ] [2 marks]

M20 concrete mix ratio:

- (A) 1:1.5:3
- (B) 1:2:4
- (C) 1:3:6
- (D) 1:1:2

Q43. [2-MCQ] [2 marks]

Taj Mahal built by:

- (A) Akbar
- (B) Shah Jahan
- (C) Aurangzeb
- (D) Humayun

Q44. [2-MCQ] [2 marks]

M20 concrete mix ratio:

- (A) 1:1.5:3
- (B) 1:2:4
- (C) 1:3:6
- (D) 1:1:2

Q45. [1-MCQ] [1 mark]

Camber on road provides:

- (A) Speed
- (B) Drainage
- (C) Aesthetics
- (D) Strength

Q46. [1-MCQ] [1 mark]

M25 concrete modulus of elasticity:

- (A) 25000 MPa
- (B) 5000 MPa
- (C) 50000 MPa
- (D) 10000 MPa

Q47. [1-NAT] [1 mark]

Slump test measures:

(Numerical Answer Type — Enter the numerical value)

Q48. [2-NAT] [2 marks]

M25 concrete modulus of elasticity:

(Numerical Answer Type — Enter the numerical value)

Q49. [2-MCQ] [2 marks]

M25 concrete modulus of elasticity:

- (A) 25000 MPa
- (B) 5000 MPa
- (C) 50000 MPa
- (D) 10000 MPa

Q50. [2-MCQ] [2 marks]

M25 concrete modulus of elasticity:

- (A) 25000 MPa
- (B) 5000 MPa
- (C) 50000 MPa
- (D) 10000 MPa

Q51. [2-MCQ] [2 marks]

Slump test measures:

- (A) Strength
 - (B) Workability
 - (C) Durability
 - (D) Setting time
-

Q52. [2-NAT] [2 marks]

Camber on road provides:

(Numerical Answer Type — Enter the numerical value)

Q53. [2-MCQ] [2 marks]

M25 concrete modulus of elasticity:

- (A) 25000 MPa
 - (B) 5000 MPa
 - (C) 50000 MPa
 - (D) 10000 MPa
-

Q54. [1-MCQ] [1 mark]

Moment distribution DF for $K_{AB}=4EI/L$, $K_{AC}=3EI/L$:

- (A) 0.57
 - (B) 0.43
 - (C) 0.5
 - (D) 0.75
-

Q55. [1-MCQ] [1 mark]

Cantilever UDL w , L . Fixed end moment:

- (A) $wL^2/2$
 - (B) wL^2
 - (C) $wL^2/4$
 - (D) $wL^2/8$
-

Q56. [1-MCQ] [1 mark]

Influence line for SS beam reaction:

- (A) Triangle
 - (B) Rectangle
 - (C) Linear decreasing
 - (D) Parabola
-

Q57. [2-NAT] [2 marks]

Cantilever UDL w , L . Fixed end moment:

(Numerical Answer Type — Enter the numerical value)

Q58. [2-NAT] [2 marks]

Moment distribution DF for $K_{AB}=4EI/L$, $K_{AC}=3EI/L$:

(Numerical Answer Type — Enter the numerical value)

Q59. [1-MCQ] [1 mark]

Sun path in northern hemisphere:

- (A) South
 - (B) North
 - (C) East
 - (D) West
-

Q60. [1-MCQ] [1 mark]

Trap in plumbing:

- (A) Prevents smell

- (B) Stores water
- (C) Both
- (D) None

Q61. [1-MCQ] [1 mark]

M20 concrete ratio:

- (A) 1:1:2
- (B) 1:1.5:3
- (C) 1:2:4
- (D) 1:3:6

Q62. [1-MCQ] [1 mark]

Zoning regulates:

- (A) Land use
- (B) Building height
- (C) Both
- (D) None

Q63. [1-MCQ] [1 mark]

Golden ratio approximately:

- (A) 1.414
- (B) 1.618
- (C) 2.0
- (D) 3.14

Q64. [1-MCQ] [1 mark]

Load-bearing wall thickness:

- (A) 23cm
- (B) 11cm
- (C) 46cm
- (D) 5cm

Q65. [1-MCQ] [1 mark]

Hardscape includes:

- (A) Plants
 - (B) Pavement
 - (C) Water
 - (D) Trees
-

ANSWER KEY

Q. No.	Type	Answer	Marks
1	1NAT	B	1
2	1MCQ	B	1
3	1MCQ	B	1
4	1MCQ	B	1
5	1MCQ	C	1
6	1MCQ	B	1
7	1MCQ	A	1
8	1MCQ	A	1
9	1MCQ	C	1
10	1MCQ	B	1
11	1MCQ	B	1
12	1MCQ	C	1
13	1MCQ	A	1

14	1MCQ	B	1
15	1MCQ	A	1
16	1MCQ	A	1
17	1MCQ	A	1
18	1MCQ	A	1
19	1MCQ	C	1
20	1MCQ	A	1
21	1MCQ	C	1
22	1NAT	B	1
23	1NAT	B	1
24	1MCQ	A	1
25	1MCQ	A	1
26	1MCQ	B	1
27	1MCQ	B	1
28	1NAT	A	1

29	1NAT	A	1
30	1NAT	A	1
31	2MCQ	A	2
32	2MCQ	C	2
33	2MCQ	A	2
34	2NAT	B	2
35	2MSQ	A	2
36	2MCQ	A	2
37	2MCQ	B	2
38	2MSQ	A	2
39	2MCQ	C	2
40	2MCQ	C	2
41	2MCQ	A	2
42	2MCQ	B	2
43	2MSQ	A	2

44	2NAT	A	2
45	2MCQ	A	2
46	2MCQ	A	2
47	2MCQ	B	2
48	2MSQ	A	2
49	2MCQ	B	2
50	2MSQ	A	2
51	2NAT	A	2
52	2MCQ	A	2
53	2MCQ	B	2
54	2NAT	A	2
55	2NAT	B	2
56	2MCQ	A	2
57	2MCQ	A	2
58	2MCQ	B	2

59	2NAT	A	2
60	2MCQ	A	2
61	2MCQ	B	2
62	2MCQ	B	2
63	2NAT	B	2
64	2MCQ	A	2
65	2MCQ	A	2

DETAILED SOLUTIONS

Q1.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Quantitative | Difficulty: easy

Q2.

Answer: B

Solution:

Correct Answer: B

Subject: Architecture | Topic: History | Difficulty: easy

Q3.

Answer: B

Solution:

Correct Answer: B

Subject: Architecture | Topic: Vernacular | Difficulty: moderate

Q4.

Answer: B

Solution:

Correct Answer: B

Subject: Building Materials | Topic: Concrete | Difficulty: moderate

Q5.

Answer: C

Solution:

Correct Answer: C

Subject: Urban Planning | Topic: Zoning | Difficulty: moderate

Q6.

Answer: B

Solution:

Correct Answer: B

Subject: Design | Topic: Principles | Difficulty: moderate

Q7.

Answer: A

Solution:

Correct Answer: A

Subject: Construction Technology | Topic: Structures | Difficulty: moderate

Q8.

Answer: A

Solution:

Correct Answer: A

Subject: Climate & Services | Topic: Solar | Difficulty: easy

Q9.

Answer: C

Solution:

Correct Answer: C

Subject: Climate & Services | Topic: Plumbing | Difficulty: difficult

Q10.

Answer: B

Solution:

Correct Answer: B

Subject: Landscape Architecture | Topic: Elements | Difficulty: moderate

Q11.

Answer: B

Solution:

Taj Mahal: built by Mughal emperor Shah Jahan (1632-1653) in memory of Mumtaz Mahal.

Subject: Architecture | Topic: Architecture | Difficulty: moderate

Q12.

Answer: C

Solution:

Gauss-Seidel converges if matrix is diagonally dominant. Jacobi needs symmetric positive definite.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q13.

Answer: A

Solution:

M20: 1:1.5:3 (cement:sand:aggregate). Characteristic strength = 20 MPa.

Subject: Architecture | Topic: Architecture | Difficulty: moderate

Q14.

Answer: B

Solution:

Camber: cross slope for rainwater drainage.

Subject: Civil Engineering | Topic: Civil Engineering | Difficulty: moderate

Q15.

Answer: A

Solution:

$DF_{AB} = 4/(4+3) = 4/7 = 0.57$.

Subject: Structural Analysis | Topic: Structural Analysis | Difficulty: moderate

Q16.

Answer: A

Solution:

$FEM = wL*(L/2) = wL^2/2$ (hogging).

Subject: Structural Analysis | Topic: Structural Analysis | Difficulty: moderate

Q17.

Answer: A

Solution:

Eigenvalues of A^2 are 1, 4, 9. Trace = sum = $1+4+9 = 14$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q18.

Answer: A

Solution:

Standard normal: $P(-1 < Z < 1)$ approx 0.68.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q19.

Answer: C

Solution:

$R_A = (L-x)/L$. Linear decreasing from L to 0.

Subject: Structural Analysis | Topic: Structural Analysis | Difficulty: easy

Q20.

Answer: A

Solution:

$E_c = 5000*\sqrt{f_{ck}} = 5000*5 = 25000$ MPa.

Subject: Civil Engineering | Topic: Civil Engineering | Difficulty: difficult

Q21.

Answer: C

Solution:

Kirchhoff: for K_n , spanning trees = $n^{(n-2)} = 4^2 = 16$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q22.

Answer: B

Solution:

$f(0)=2$, $f(1)=0$, $f(-1)=4$, $f(2)=4$. $f(-2)=-8+6+2=0$. Roots at $x=1$ (double?) and $x=-2$. $f(x)=(x-1)^2(x+2)$. Two distinct real roots.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q23.

Answer: B

Solution:

Slump: workability/consistency of fresh concrete.

Subject: Civil Engineering | Topic: Civil Engineering | Difficulty: moderate

Q24.

Answer: A

Solution:

$L\{e^{(-at)}\} = 1/(s+a)$. Here $a=3$: $1/(s+3)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q25.

Answer: A

Solution:

Homogeneous ODE. Substituting $y=x*v$ and $y(1)=0$ gives $y=0$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q26.

Answer: B

Solution:

$(1+1/n)^{2n} = [(1+1/n)^n]^2 \rightarrow e^2$ as $n \rightarrow \infty$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q27.

Answer: B

Solution:

$|x|$ is continuous everywhere. Left derivative at 0 = -1, right = +1. Not differentiable at 0.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q28.

Answer: A

Solution:

Area = πr^2 . Circumference = $2\pi r$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q29.

Answer: A

Solution:

$f(x)=x$ is odd on symmetric interval. Fourier series has only sine terms.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q30.

Answer: A

Solution:

$P(X \geq 2) = 1 - e^{-4}(1+4) = 1 - 5e^{-4} \approx 0.908$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q31.

Answer: A

Solution:

$f(i) = i^2 + 3i + 2 = -1 + 3i$. $\text{Re}(f(i)) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q32.

Answer: C

Solution:

$y = A\cos(2x) + B\sin(2x)$. $A=1$, $B=0$. $y(\pi/2) = \cos(\pi) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q33.

Answer: A

Solution:

$|\langle u, v \rangle| \leq \|u\| \cdot \|v\|$. Equality when u and v are linearly dependent.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q34.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = C \cdot x$. Wait: $y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = C \cdot x$. That's option A. Hmm, let me check: $y \, dx - x \, dy = 0 \Rightarrow y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow y = C \cdot x$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q35.

Answer: A

Solution:

Poisson: mean = lambda = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q36.

Answer: A

Solution:

Row 2 = 2*Row 1, Row 3 = 3*Row 1. All rows linearly dependent. Rank = 1.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q37.

Answer: B

Solution:

$\text{proj} = ((v \cdot w)/(w \cdot w)) \cdot w = (4/5) \cdot (2, 1, 0) = (8/5, 4/5, 0)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q38.

Answer: A

Solution:

$L\{f(t)\} = -d/ds(F(s))$. $F(s) = 1/(s+2)$. $-d/ds[1/(s+2)] = 1/(s+2)^2$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q39.

Answer: C

Solution:

Rolle's: f continuous on $[a, b]$, differentiable on (a, b) , $f(a) = f(b)$. Then $f'(c) = 0$ for some c in (a, b) .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q40.

Answer: C

Solution:

$d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2 \tan x$. At $\pi/4$: $1 + 2 = 3$. Hmm. Let me recalc: $d/dx \ln(\sec x) = \tan x$. $d/dx \ln(\tan x) = \sec^2 x / \tan x = 1/(\sin x \cos x) = 2 \csc(2x)$. At $\pi/4$: $1 + 2 = 3$. Not in options. The derivative of $\ln(\sec x \cdot \tan x) = d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2 \tan x$. At $\pi/4$: $1 + 2 = 3$. Hmm.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q41.

Answer: A

Solution:

PGF of Poisson(lambda): $G(z) = E[z^X] = \exp(\lambda(z-1))$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q42.

Answer: B

Solution:

$P(A|B) = P(A \cap B)/P(B) = 0.2/0.4 = 0.5$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q43.

Answer: A

Solution:

Eigenvalues are 3 and 1. $[1,0]^T$ is eigenvector for eigenvalue 1. $A^{50}[1,0]^T = [1,0]^T$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q44.

Answer: A

Solution:

$E_c = 5000 \cdot \sqrt{f_{ck}} = 5000 \cdot 5 = 25000 \text{ MPa}$.

Subject: Civil Engineering | Topic: Civil Engineering | Difficulty: moderate

Q45.

Answer: A

Solution:

M20: 1:1.5:3 (cement:sand:aggregate). Characteristic strength = 20 MPa.

Subject: Architecture | Topic: Architecture | Difficulty: moderate

Q46.

Answer: A

Solution:

$f(x)=x$ is odd on symmetric interval. Fourier series has only sine terms.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q47.

Answer: B

Solution:

$(1+1/n)^{2n} = [(1+1/n)^n]^2 \rightarrow e^2$ as $n \rightarrow \infty$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q48.

Answer: A

Solution:

$P(X \geq 2) = 1 - e^{-4(1+4)} = 1 - 5e^{-4} \approx 0.908$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q49.

Answer: B

Solution:

Taj Mahal: built by Mughal emperor Shah Jahan (1632-1653) in memory of Mumtaz Mahal.

Subject: Architecture | Topic: Architecture | Difficulty: difficult

Q50.

Answer: A

Solution:

$P(X \geq 2) = 1 - e^{-4(1+4)} = 1 - 5e^{-4} \approx 0.908$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q51.

Answer: A

Solution:

Row 2 = 2*Row 1, Row 3 = 3*Row 1. All rows linearly dependent. Rank = 1.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q52.

Answer: A

Solution:

$E_c = 5000 \cdot \sqrt{f_{ck}} = 5000 \cdot 5 = 25000 \text{ MPa}$.

Subject: Civil Engineering | Topic: Civil Engineering | Difficulty: easy

Q53.

Answer: B

Solution:

$P(A|B) = P(A \cap B)/P(B) = 0.2/0.4 = 0.5$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q54.

Answer: A

Solution:

$FEM = wL \cdot (L/2) = wL^2/2$ (hogging).

Subject: Structural Analysis | Topic: Structural Analysis | Difficulty: easy

Q55.

Answer: B

Solution:

Taj Mahal: built by Mughal emperor Shah Jahan (1632-1653) in memory of Mumtaz Mahal.

Subject: Architecture | Topic: Architecture | Difficulty: easy

Q56.

Answer: A

Solution:

$E_c = 5000 \cdot \sqrt{f_{ck}} = 5000 \cdot 5 = 25000 \text{ MPa}$.

Subject: Civil Engineering | Topic: Civil Engineering | Difficulty: moderate

Q57.

Answer: A

Solution:

M20: 1:1.5:3 (cement:sand:aggregate). Characteristic strength = 20 MPa.

Subject: Architecture | Topic: Architecture | Difficulty: moderate

Q58.

Answer: B

Solution:

Slump: workability/consistency of fresh concrete.

Subject: Civil Engineering | Topic: Civil Engineering | Difficulty: difficult

Q59.

Answer: A

Solution:

$DF_{AB} = 4/(4+3) = 4/7 = 0.57$.

Subject: Structural Analysis | Topic: Structural Analysis | Difficulty: easy

Q60.

Answer: A

Solution:

M20: 1:1.5:3 (cement:sand:aggregate). Characteristic strength = 20 MPa.

Subject: Architecture | Topic: Architecture | Difficulty: easy

Q61.

Answer: B

Solution:

Taj Mahal: built by Mughal emperor Shah Jahan (1632-1653) in memory of Mumtaz Mahal.

Subject: Architecture | Topic: Architecture | Difficulty: easy

Q62.

Answer: B

Solution:

$f(0)=2$, $f(1)=0$, $f(-1)=4$, $f(2)=4$. $f(-2)=-8+6+2=0$. Roots at $x=1$ (double?) and $x=-2$. $f(x)=(x-1)^2(x+2)$. Two distinct real roots.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q63.

Answer: B

Solution:

Camber: cross slope for rainwater drainage.

Subject: Civil Engineering | Topic: Civil Engineering | Difficulty: easy

Q64.

Answer: A

Solution:

M20: 1:1.5:3 (cement:sand:aggregate). Characteristic strength = 20 MPa.

Subject: Architecture | Topic: Architecture | Difficulty: easy

Q65.

Answer: A

Solution:

$$E_c = 5000 \sqrt{f_{ck}} = 5000 \sqrt{5} = 25000 \text{ MPa.}$$

Subject: Civil Engineering | Topic: Civil Engineering | Difficulty: moderate

PAPER BLUEPRINT

Subject Distribution:

Engineering Mathematics: 32 questions, 52 marks
Architecture: 11 questions, 18 marks
Civil Engineering: 9 questions, 15 marks
Structural Analysis: 5 questions, 7 marks
Climate & Services: 2 questions, 2 marks
General Aptitude: 1 questions, 1 marks
Building Materials: 1 questions, 1 marks
Urban Planning: 1 questions, 1 marks
Design: 1 questions, 1 marks
Construction Technology: 1 questions, 1 marks
Landscape Architecture: 1 questions, 1 marks

Difficulty Distribution:

Easy: 22 | Moderate: 34 | Difficult: 9

Statistics:

Total Questions: 65
Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.